Model Order Reduction

Parametric PDE

Consider the FEM discretization of the Poisson problem on Ω_h: find $u_h \in V_h$ such that

$$a(u_h, v_h) = F(v_h) \qquad \forall v_h \in V_h, \qquad +BC$$

and suppose that there are some parameters $\mu \in \mathbb{P} \subset \mathbb{R}^P$ that modify the problem:

- $\Omega_h(\mu)$
- $a(u_h, v_h; \mu)$
- $F(v_b; \mu)$
- BC(μ)

We would like to

- know the solution for many μ_i (e.g. we want to compute statistics, we need to do optimization)
- we need a fast evaluation $\mu \mapsto u_h(\mu)$.

But it is **too expensive** to compute all FEM simulations

Model order reduction (MOR)

(or reduced order model (ROM))

We look for a smaller space $V_n = \langle \{\psi_i\}_{i=1}^n \rangle$ with dimension $n \ll N_h$, which is a **good approximation** of the space generated by $\langle \{u_h(\mu) \text{ with } \mu \in \mathbb{P}\} \rangle \subset V_h$, where $\mathbb{P}$ is the parameter space.

1. What does the problem becomes in the space V_n?
2. Is solving the problem in V_n faster than in V_h?
3. How do I find the space V_n?

- FEM: $V_h = \langle \{\varphi_i\}_{i=1}^{N_h} \rangle$ and $u_h(x) := \sum_{j=1}^{N_h} \varphi_j(x) u_j$ with u_j the unknown coefficients.

$$\sum_j a(\varphi_i, \varphi_j; \mu) u_j = F(\varphi_i; \mu) \qquad \forall i = 1, \dots, N_h$$

- ROM Galerkin Projection: $V_n = \langle \{\psi_i\}_{i=1}^{n} \rangle$ and $u_n(x) := \sum_{j=1}^{n} \psi_j(x) \hat{u}_j$ with u_j the unknown coefficients.

$$\sum_j a(\psi_i, \psi_j; \mu) \hat{u}_j = F(\psi_i; \mu) \qquad \forall i = 1, \dots, n.$$

Matrix formulation

$\psi_i(x) = \sum_{j=1}^{N_h} B_{ji} \varphi_j(x)$. So, we can look at the ROM basis functions as a set of coefficients $B \in \mathbb{R}^{N_h \times n}$ that define the ROM space V_n as a subspace of V_h.

The FEM problem is given by $A(\mu) u = F(\mu)$, where $A_{ij}(\mu) = a(\varphi_i, \varphi_j; \mu)$ and $F_i(\mu) = F(\varphi_i; \mu)$.

The ROM problem is given by $\hat{A}(\mu) \hat{u} = \hat{F}(\mu)$, where

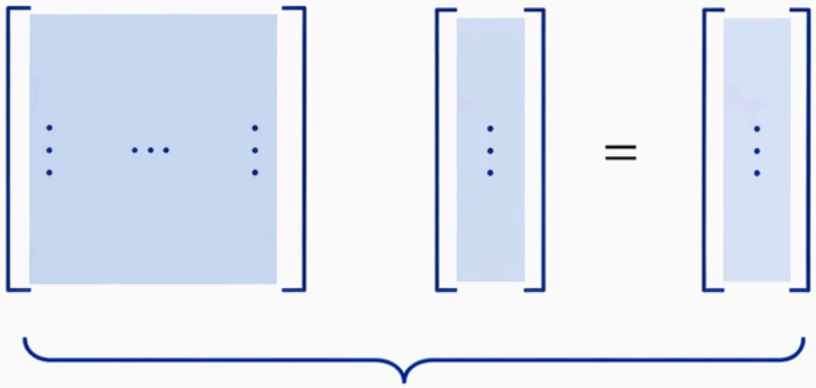
$$\hat{A}_{ij}(\mu) = a(\psi_i, \psi_j; \mu) = \sum_{k,l=1}^{N_h} a(B_{ki} \varphi_k, B_{lj} \varphi_l; \mu) = \sum_{k,l=1}^{N_h} B_{ki}(\mu) A_{kl}(\mu) B_{lj}(\mu) = (B^T A(\mu) B)_{ij}$$

and $\hat{F}_i(\mu) = F(\psi_i; \mu) = \sum_{j=1}^{N_h} B_{ji}(\mu) F_j(\mu) = (B^T F(\mu))_i$.

Full-Order Finite Element Model

$$Au = F$$

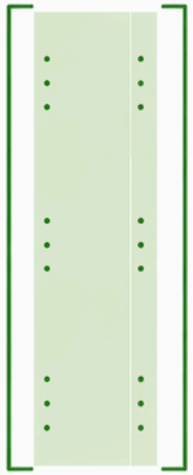
$A \in \mathbb{R}^{N_h \times N_h}$ $\quad u \in \mathbb{R}^{N_h \times 1}$ $\quad F \in \mathbb{R}^{N_h \times 1}$



N_h = dimension of the FE space (large)

Reduction Basis

$B \in \mathbb{R}^{N_h \times n}$



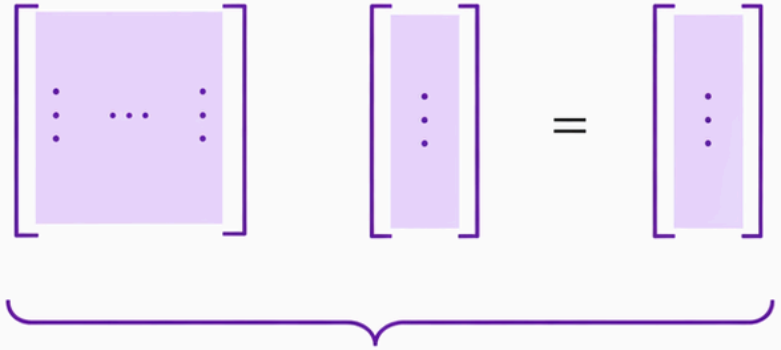
$N_h \gg n$
(tall and skinny)

Reduced-Order Model via Projection

Project full system:

$$B^{\mathrm{T}}AB = B^{\mathrm{T}}F$$

$B^{\mathrm{T}}AB \in \mathbb{R}^{n \times n}$ $\quad \hat{u} \in \mathbb{R}^{n \times 1}$ $\quad B^{\mathrm{T}}F \in \mathbb{R}^{n \times 1}$



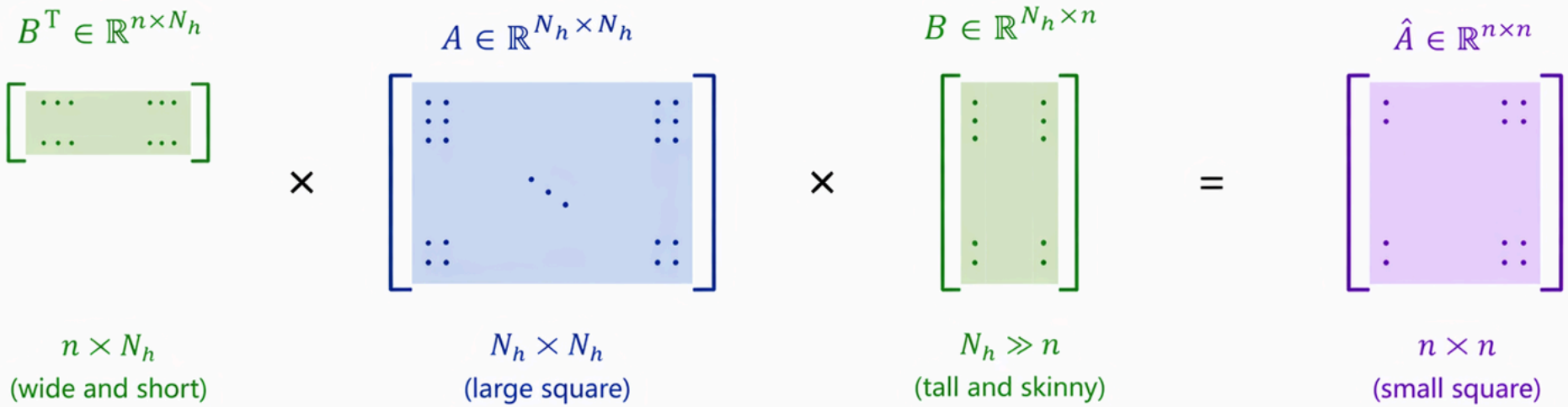
n = dimension of the reduced space (small)

Galerkin Projection

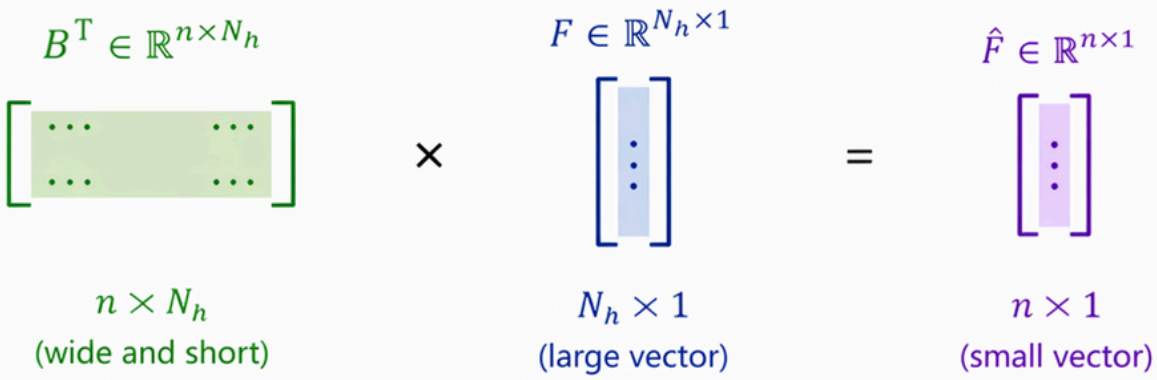
$$\hat{A} = B^{\mathrm{T}}AB \qquad \hat{F} = B^{\mathrm{T}}F \qquad \hat{u} = B^{\mathrm{T}}u$$

with $B^{\mathrm{T}}B = I_n$ (orthonormal basis)

Matrix Projection: $\hat{A} = B^{\mathrm{T}} A B$



Vector Projection: $\hat{F} = B^{\mathrm{T}} F$



Affine decomposition of the system

The ROM problem is really cheaper if we can also assemble the ROM matrices ahead of time in a **offline** phase, i.e. if we can write

$$A(\mu) = \sum_{q=1}^{Q_a} \Theta_q^a(\mu) A_q, \qquad F(\mu) = \sum_{q=1}^{Q_F} \theta_q^F(\mu) F_q,$$

where $A_q \in \mathbb{R}^{N_h \times N_h}$ and $F_q \in \mathbb{R}^{N_h}$ are parameter-independent matrices and vectors and $\Theta_q^a(\mu) : \mathbb{P} \to \mathbb{R}$ and $\theta_q^F(\mu) : \mathbb{P} \to \mathbb{R}$ are parameter-dependent scalar functions.
In this way, we can precompute the ROM matrices and vectors

$$\hat{A}_q = B^T A_q B, \qquad \hat{F}_q = B^T F_q$$

and then assemble the ROM system in the **online** phase as

$$\hat{A}(\mu) = \sum_{q=1}^{Q_a} \Theta_q^a(\mu) \hat{A}_q = \sum_{q=1}^{Q_a} \Theta_q^a(\mu) B^T A_q B = B^T A(\mu) B,$$

$$\hat{F}(\mu) = \sum_{q=1}^{Q_F} \theta_q^F(\mu) \hat{F}_q = \sum_{q=1}^{Q_F} \theta_q^F(\mu) B^T F_q = B^T F(\mu).$$

Example of affine decomposition: the Poisson problem with a parameter-dependent diffusion coefficient

Consider the Poisson problem with a parameter-dependent diffusion coefficient $\kappa(\mu)$:

$$\begin{cases} -\nabla \cdot (\kappa(x;\mu)\nabla u) = f & \text{in } \Omega, \\ u(x) = 0, & x \in \Gamma_{top}, \\ \kappa(x;\mu)\nabla u(x) \cdot \mathbf{n} = 0, & x \in \Gamma_{side}, \\ \kappa(x;\mu)\nabla u(x) \cdot \mathbf{n} = \mu_2, & x \in \Gamma_{base}. \end{cases}$$

Where the diffusion coefficient $\kappa(\mu)$ is given by

$$\kappa(\mu) = \begin{cases} \mu_1 & \text{in } \Omega_1, \\ 1 & \text{in } \Omega_2. \end{cases}$$

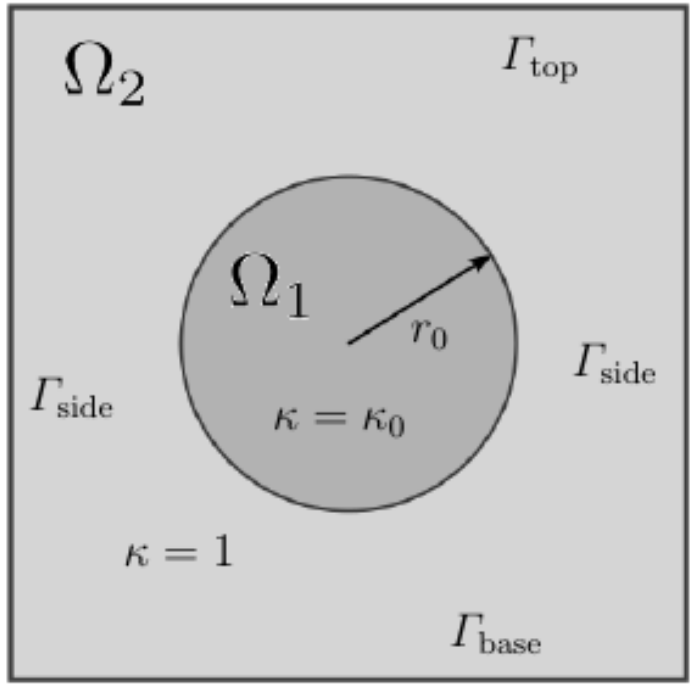
The affine decomposition of a can be written as

$$a(\varphi_i, \varphi_j; \mu) = \sum_{q=1}^{2} \Theta_q^a(\mu) a_q(\varphi_i, \varphi_j),$$

$$a_1(\varphi_i, \varphi_j) = \int_{\Omega_1} \nabla\varphi_i \cdot \nabla\varphi_j dx, \qquad \Theta_1^a(\mu) = \mu_1,$$

$$a_2(\varphi_i, \varphi_j) = \int_{\Omega_2} \nabla\varphi_i \cdot \nabla\varphi_j dx, \qquad \Theta_2^a(\mu) = \mu_2,$$

$$f_1(\varphi_i) = \int_{\Gamma_{base}} \varphi_i ds, \qquad \theta_1^F(\mu) = \mu_2.$$



How to build the ROM space V_n? Proper orthogonal decomposition (POD)

In an **offline** phase, we compute the solution of the full order model (FOM) for a set of parameters $\{\mu_1, \ldots, \mu_{N_s}\}$ and we collect the solutions in a matrix $S = [u_h(\mu_1), \ldots, u_h(\mu_{N_s})] \in \mathbb{R}^{N_h \times N_s}$, called **snapshot matrix** ($N_s \ll N_h$).

We then apply the **SVD** algorithm to the snapshot matrix S and we select the first n left singular vectors as the ROM basis functions $\{\psi_i\}_{i=1}^n$.

$$S = U\Sigma V^T, \qquad U = [\psi_1, \ldots, \psi_{N_s}] \in \mathbb{R}^{N_h \times N_s}, \qquad \Sigma = diag(\sigma_1, \ldots, \sigma_{N_s}) \in \mathbb{R}^{N_s \times N_s}, \qquad V = [v_1, \ldots, v_{N_s}] \in \mathbb{R}^{N_s \times N_s},$$

with $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_{N_s} \geq 0$ the singular values of S.

Keep only the first n singular values, which gives the most important information of the snapshot matrix, and discard the rest.
So, we approximate

$$S \approx \hat{S} = \hat{U}\hat{\Sigma}\hat{V}^T, \qquad \hat{U} = [\psi_1, \ldots, \psi_n] \in \mathbb{R}^{N_h \times n}, \qquad \hat{\Sigma} = diag(\sigma_1, \ldots, \sigma_n) \in \mathbb{R}^{n \times n}, \qquad \hat{V} = [v_1, \ldots, v_n] \in \mathbb{R}^{N_s \times n}.$$

The ROM basis functions are given by $\psi_i = B_{:,i} := U_{:,i}$ for $i = 1, \ldots, n$.

Given data matrix

$S \in \mathbb{R}^{N_h \times N_s}$

Singular Value Decomposition (SVD)

$$\boxed{S = U \Sigma V^T}$$

$$\underset{N_s}{\overset{S}{N_h \begin{bmatrix} \vdots & \cdots & \vdots \end{bmatrix}}} = \underset{N_s}{\overset{U \in \mathbb{R}^{N_h \times N_s}}{N_h \begin{bmatrix} \vdots & \cdots & \vdots \end{bmatrix}}} \times \underset{N_s}{\overset{\Sigma \in \mathbb{R}^{N_s \times N_s}}{\begin{bmatrix} \sigma_1 & & & 0 \\ & \sigma_2 & & \\ & & \ddots & \\ 0 & & & \sigma_{N_s} \end{bmatrix}}} \times \underset{N_s}{\overset{V^T \in \mathbb{R}^{N_s \times N_s}}{\begin{bmatrix} \vdots & \vdots \end{bmatrix}}}$$

Dimensions:

- $U \in \mathbb{R}^{N_h \times N_s}$
- $\Sigma \in \mathbb{R}^{N_s \times N_s}$
- $V^T \in \mathbb{R}^{N_s \times N_s}$

(with $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{N_s} \geq 0$)

Truncated SVD (rank–n approximation)

Truncate to retain only the first n singular values

$(n \ll N_s)$

$$\boxed{\hat{S} = \hat{U} \hat{\Sigma} \hat{V}^T}$$

$$\underset{n}{\overset{\hat{S} \in \mathbb{R}^{N_h \times n}}{N_h \begin{bmatrix} \vdots & \vdots \end{bmatrix}}} = \underset{n}{\overset{\hat{U} \in \mathbb{R}^{N_h \times n}}{N_h \begin{bmatrix} \vdots & \vdots \end{bmatrix}}} \times \underset{n}{\overset{\hat{\Sigma} \in \mathbb{R}^{n \times n}}{\begin{bmatrix} \sigma_1 & & & 0 \\ & \sigma_2 & & \\ & & \ddots & \\ 0 & & & \sigma_n \end{bmatrix}}} \times \underset{n}{\overset{\hat{V}^T \in \mathbb{R}^{n \times n}}{\begin{bmatrix} \vdots & \vdots \end{bmatrix}}}$$

Dimensions after truncation:

- $\hat{U} \in \mathbb{R}^{N_h \times n}$
- $\hat{\Sigma} \in \mathbb{R}^{n \times n}$
- $\hat{V}^T \in \mathbb{R}^{n \times n}$

(keeps the first n singular values)

Best rank–n approximation: $\hat{S} = \underset{X \in \mathbb{R}^{N_h \times n}}{\operatorname{argmin}} \; \|S - X\|_F$ (Eckart–Young–Mirsky theorem)